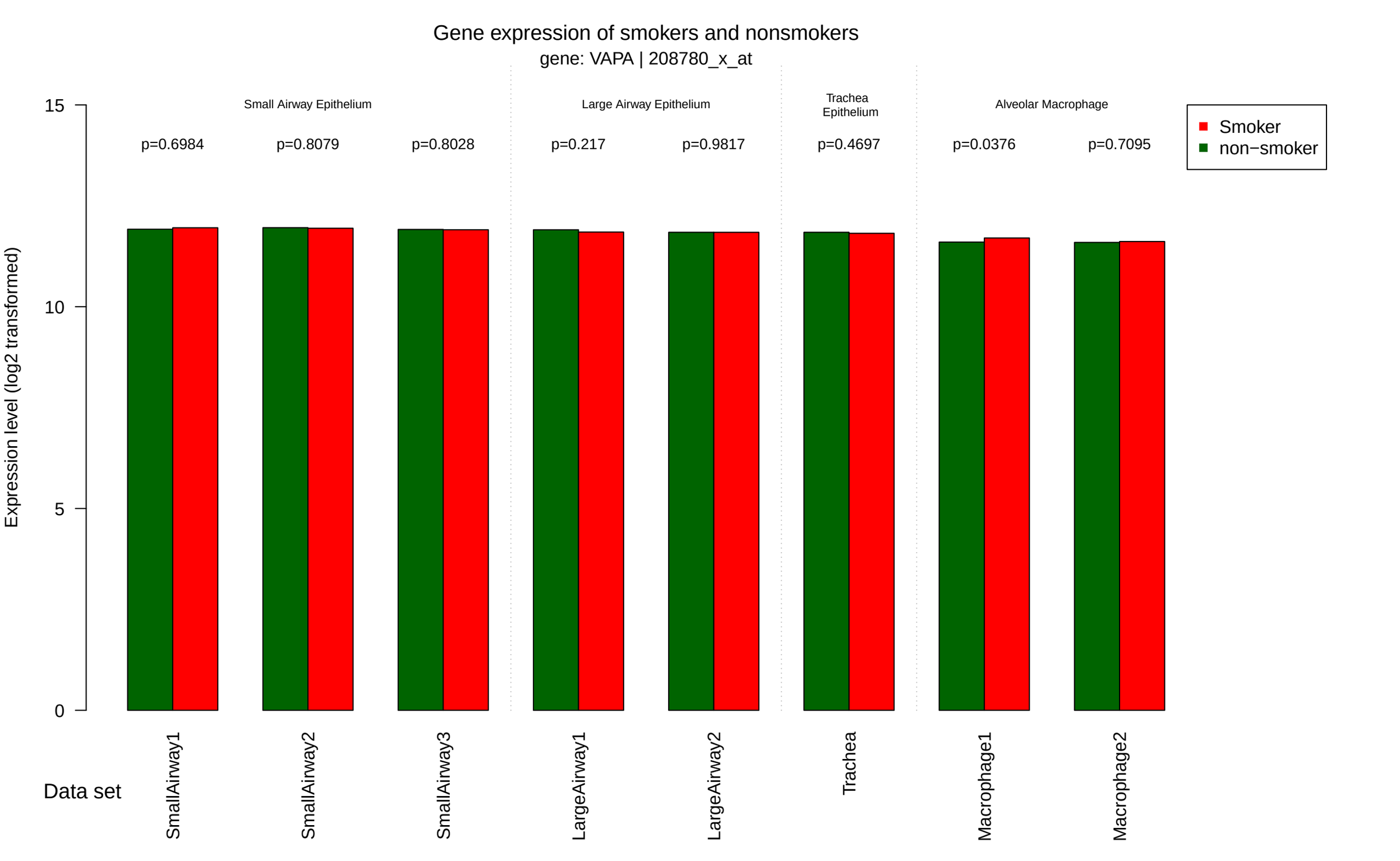
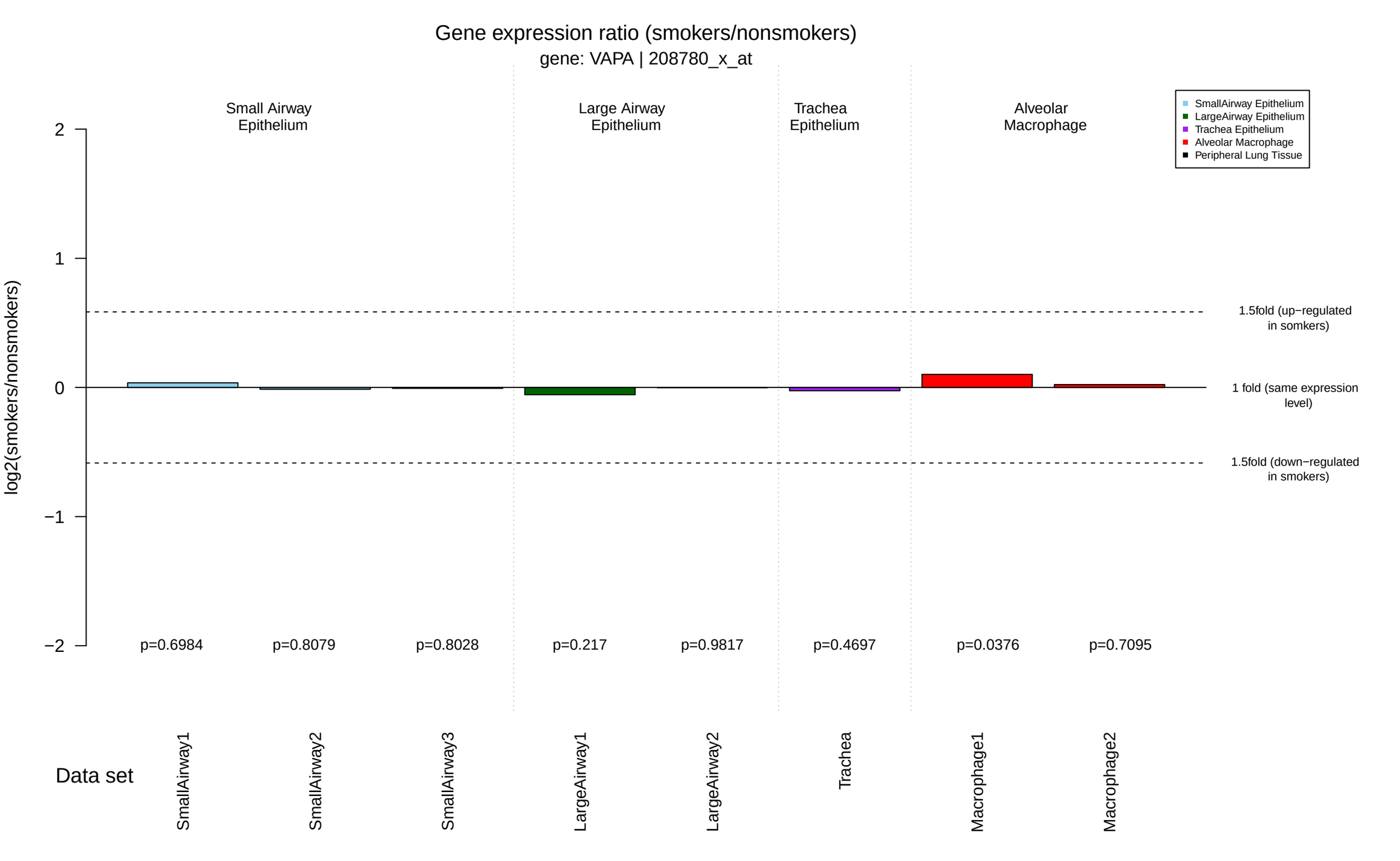
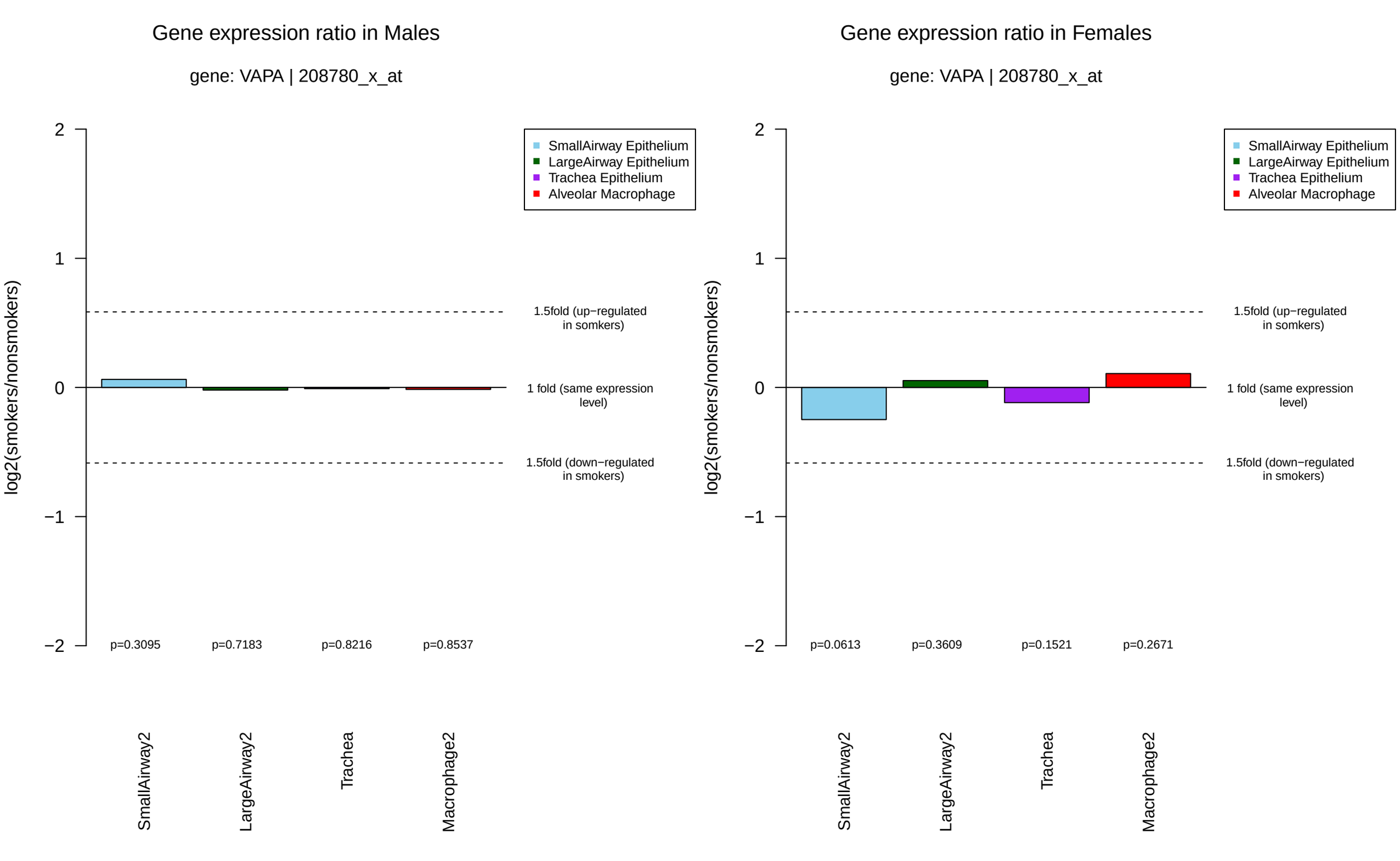
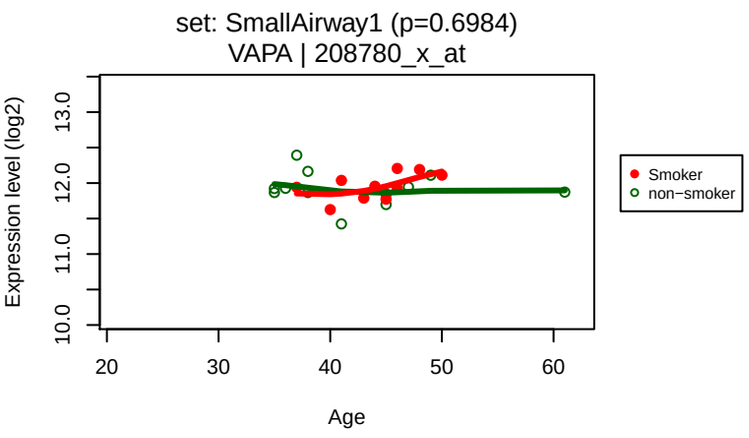
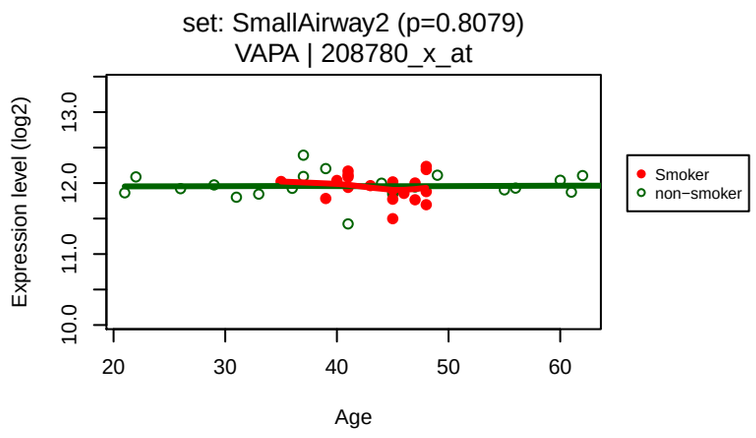
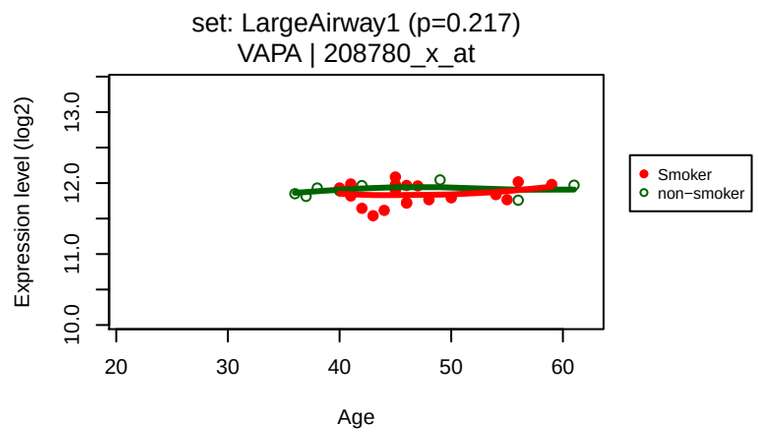

Gene expression of smokers and nonsmokers
gene: VAPA | 208780_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6984
p=0.8079
p=0.8028
p=0.217
p=0.9817
p=0.4697
p=0.0376
p=0.7095
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

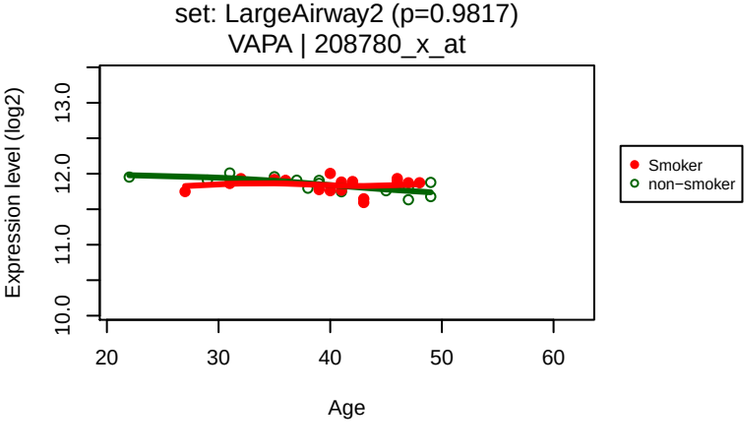

Gene expression ratio (smokers/nonsmokers)
gene: VAPA | 208780_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6984
p=0.8079
p=0.8028
p=0.217
p=0.9817
p=0.4697
p=0.0376
p=0.7095
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

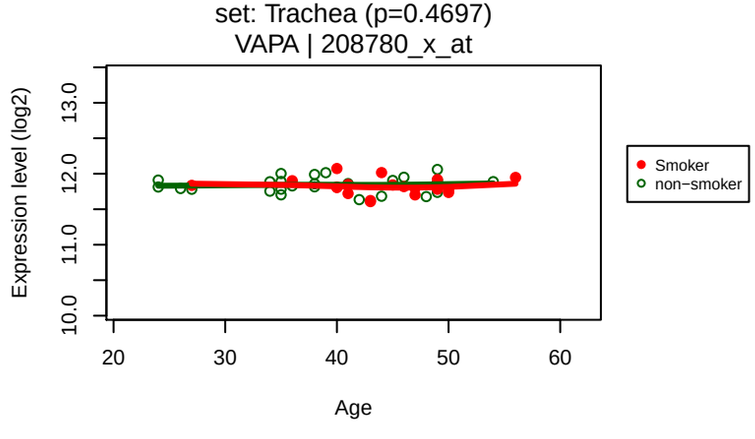

Gene expression ratio in Males
gene: VAPA | 208780_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3095
p=0.7183
p=0.8216
p=0.8537
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VAPA | 208780_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0613
p=0.3609
p=0.1521
p=0.2671
SmallAirway2
LargeAirway2
Trachea
Macrophage2

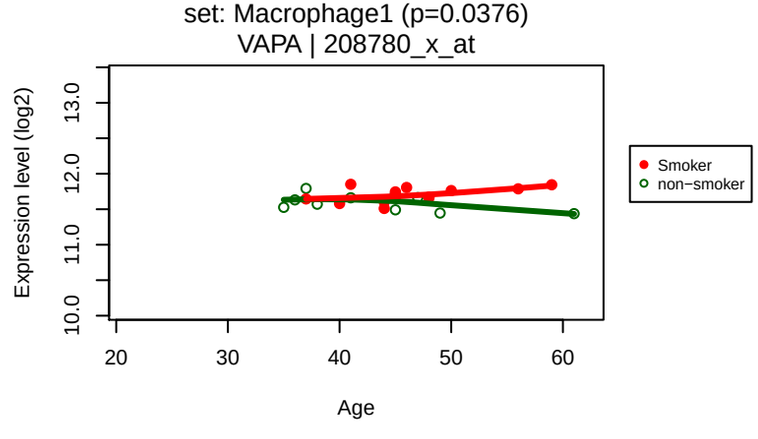

set: SmallAirway1 (p=0.6984)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker

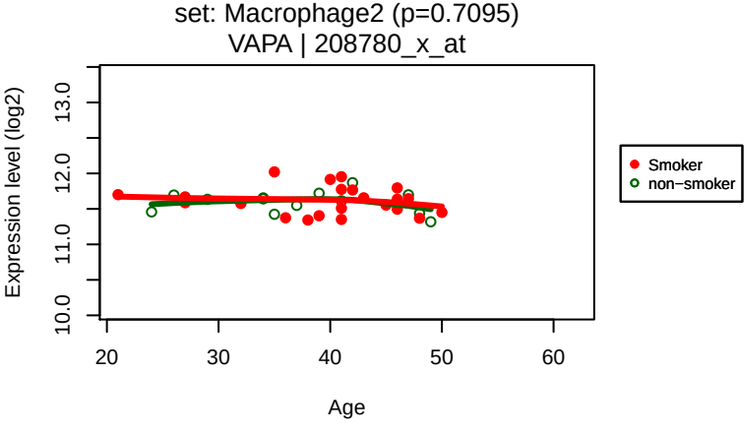

set: SmallAirway2 (p=0.8079)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker

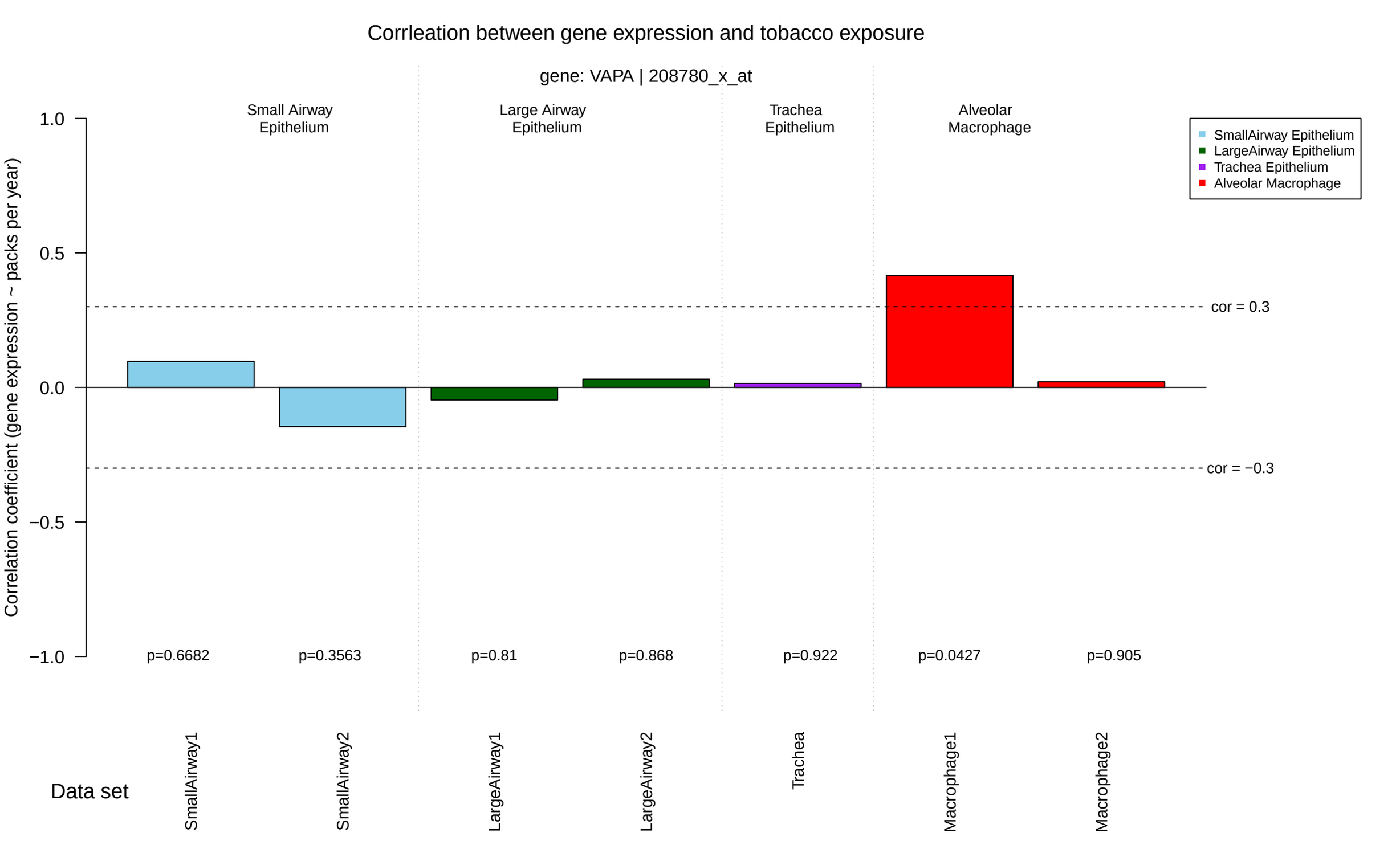

set: LargeAirway1 (p=0.217)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9817)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4697)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0376)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7095)
VAPA | 208780_x_at
Expression level (log2)
10.0
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VAPA | 208780_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6682
p=0.3563
p=0.81
p=0.868
p=0.922
p=0.0427
p=0.905
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2